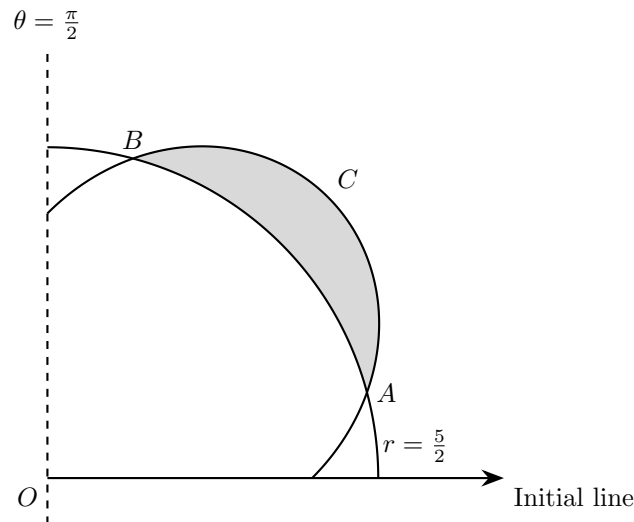


## Questions

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1. The diagram shows part of the curve  $C$  with polar equation

$$r = 2 + \sin 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

and the circle

$$r = \frac{5}{2}$$

The curve and the circle intersect at the points  $A$  and  $B$ , as shown.

Find, correct to three significant figures, the area of the shaded region enclosed by  $C$  and the circle.

[9]

### Solution

For the intersection points of the curve and the circle, set the two values of  $r$  equal:

$$2 + \sin 2\theta = \frac{5}{2}$$

So

$$\sin 2\theta = \frac{1}{2}$$

Since  $0 \leq \theta \leq \frac{\pi}{2}$ , we have  $0 \leq 2\theta \leq \pi$ , so the solutions are

$$2\theta = \frac{\pi}{6}, \quad \frac{5\pi}{6}$$

Hence

$$\theta = \frac{\pi}{12}, \quad \frac{5\pi}{12}$$

So the points of intersection are at  $\theta = \frac{\pi}{12}$  and  $\theta = \frac{5\pi}{12}$ .

For  $\frac{\pi}{12} \leq \theta \leq \frac{5\pi}{12}$ , we have  $\sin 2\theta \geq \frac{1}{2}$ , so

$$2 + \sin 2\theta \geq \frac{5}{2}$$

Therefore the curve  $C$  lies outside the circle on this interval, and the required area is

$$A = \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left[ (2 + \sin 2\theta)^2 - \left( \frac{5}{2} \right)^2 \right] d\theta$$

Now expand the integrand:

$$\begin{aligned}
 A &= \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left( 4 + 4 \sin 2\theta + \sin^2 2\theta - \frac{25}{4} \right) d\theta \\
 &= \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left( -\frac{9}{4} + 4 \sin 2\theta + \sin^2 2\theta \right) d\theta
 \end{aligned}$$

Use the identity

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

Then

$$\begin{aligned}
 A &= \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left( -\frac{9}{4} + 4 \sin 2\theta + \frac{1 - \cos 4\theta}{2} \right) d\theta \\
 &= \frac{1}{2} \int_{\pi/12}^{5\pi/12} \left( -\frac{7}{4} + 4 \sin 2\theta - \frac{1}{2} \cos 4\theta \right) d\theta
 \end{aligned}$$

Integrating,

$$A = \left[ -\frac{7\theta}{8} - \cos 2\theta - \frac{1}{16} \sin 4\theta \right]_{\pi/12}^{5\pi/12}$$

Now substitute the limits:

$$A = \left( -\frac{35\pi}{96} - \cos \frac{5\pi}{6} - \frac{1}{16} \sin \frac{5\pi}{3} \right) - \left( -\frac{7\pi}{96} - \cos \frac{\pi}{6} - \frac{1}{16} \sin \frac{\pi}{3} \right)$$

Using

$$\cos \frac{5\pi}{6} = -\frac{\sqrt{3}}{2}, \quad \sin \frac{5\pi}{3} = -\frac{\sqrt{3}}{2}, \quad \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}, \quad \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

gives

$$\begin{aligned}
 A &= \left( -\frac{35\pi}{96} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{32} \right) - \left( -\frac{7\pi}{96} - \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{32} \right) \\
 &= -\frac{28\pi}{96} + \frac{34\sqrt{3}}{32} \\
 &= -\frac{7\pi}{24} + \frac{17\sqrt{3}}{16}
 \end{aligned}$$

So the exact area is

$$A = \frac{17\sqrt{3}}{16} - \frac{7\pi}{24}$$

and numerically

$$A \approx 0.924$$

Therefore, the area of the shaded region is 0.924 square units, correct to three significant figures.

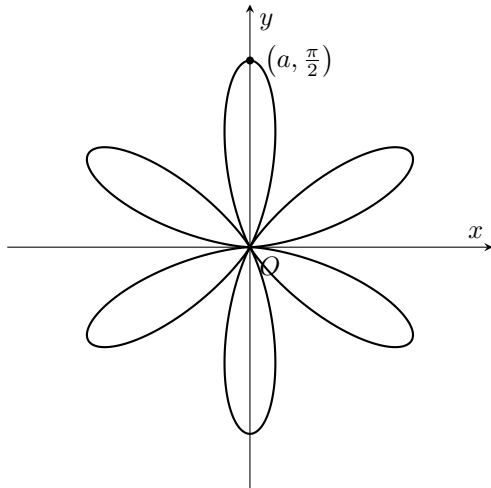
2. A curve has polar equation  $r = a \sin^2 3\theta$ , where  $a > 0$  and  $0 \leq \theta < 2\pi$ .

(a) Sketch the curve. [2]

(b) Find, in exact form, the total area of the regions enclosed by the curve. [7]

### Solution

(a) The sketch should look like:



Since  $r = a \sin^2 3\theta \geq 0$  for all  $\theta$ , every  $\theta$  gives a point on the curve. It passes through the pole when  $\sin 3\theta = 0$ , that is at  $\theta = \frac{n\pi}{3}$  for  $n = 0, 1, \dots, 5$ , so the curve is a rose with 6 petals, one between each pair of consecutive zeros. The maximum value of  $r$  is  $a$ , when  $\sin^2 3\theta = 1$ , that is at the petal tips  $\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$ .

(b) The polar area formula is

$$A = \frac{1}{2} \int r^2 d\theta$$

One petal is traced out as  $\theta$  goes from 0 to  $\frac{\pi}{3}$ , since  $r = 0$  at both endpoints and  $r > 0$  between them. So the area of one petal is

$$A_1 = \frac{1}{2} \int_0^{\pi/3} a^2 \sin^4(3\theta) d\theta$$

Use

$$\sin^4 x = \left( \frac{1 - \cos 2x}{2} \right)^2 = \frac{1 - 2\cos 2x + \cos^2 2x}{4} = \frac{1 - 2\cos 2x + \frac{1 + \cos 4x}{2}}{4} = \frac{3 - 4\cos 2x + \cos 4x}{8}$$

Therefore

$$\sin^4(3\theta) = \frac{3 - 4\cos 6\theta + \cos 12\theta}{8}$$

Substitute into the integral:

$$\begin{aligned} A_1 &= \frac{a^2}{2} \int_0^{\pi/3} \frac{3 - 4\cos 6\theta + \cos 12\theta}{8} d\theta \\ &= \frac{a^2}{16} \int_0^{\pi/3} (3 - 4\cos 6\theta + \cos 12\theta) d\theta \\ &= \frac{a^2}{16} \left[ 3\theta - \frac{4}{6} \sin 6\theta + \frac{1}{12} \sin 12\theta \right]_0^{\pi/3} \end{aligned}$$

Now evaluate:

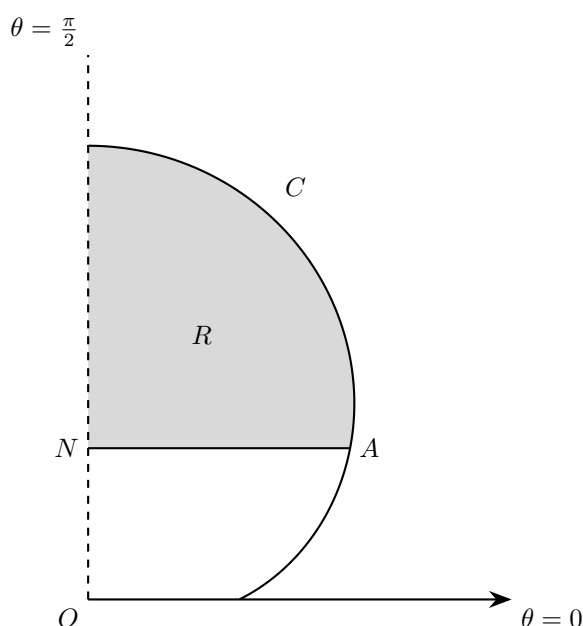
$$\sin\left(6 \cdot \frac{\pi}{3}\right) = \sin 2\pi = 0, \quad \sin\left(12 \cdot \frac{\pi}{3}\right) = \sin 4\pi = 0$$

Hence

$$\begin{aligned} A_1 &= \frac{a^2}{16} \left[ 3 \cdot \frac{\pi}{3} - 0 + 0 \right] \\ &= \frac{\pi a^2}{16} \end{aligned}$$

There are 6 identical petals, so the total area is

$$\begin{aligned} A_{\text{total}} &= 6 \cdot \frac{\pi a^2}{16} \\ &= \frac{3\pi a^2}{8} \end{aligned}$$



3. The curve  $C$  shown in the diagram above has polar equation

$$r = 1 + 2 \sin \theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

The point  $A$  on  $C$  has  $y$ -coordinate 1.

The point  $N$  lies on the line  $\theta = \frac{\pi}{2}$  and  $AN$  is perpendicular to the line  $\theta = \frac{\pi}{2}$ .

The finite region  $R$ , shown shaded in the diagram above, is bounded by the curve  $C$ , the line  $\theta = \frac{\pi}{2}$  and the line  $AN$ .

Find the exact area of the shaded region  $R$ .

[9]

### Solution

For the curve

$$r = 1 + 2 \sin \theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

the point  $A$  has  $y$ -coordinate 1.

In polar coordinates,

$$y = r \sin \theta$$

so at  $A$ ,

$$(1 + 2 \sin \theta) \sin \theta = 1$$

Hence

$$\begin{aligned} 2 \sin^2 \theta + \sin \theta - 1 &= 0 \\ (2 \sin \theta - 1)(\sin \theta + 1) &= 0 \end{aligned}$$

Since  $0 \leq \theta \leq \frac{\pi}{2}$ , we take

$$\sin \theta = \frac{1}{2}$$

so

$$\theta_A = \frac{\pi}{6}$$

Then

$$r_A = 1 + 2 \sin \frac{\pi}{6} = 1 + 2 \left( \frac{1}{2} \right) = 2$$

Therefore

$$A = \left(2, \frac{\pi}{6}\right)$$

and in Cartesian coordinates

$$x = 2 \cos \frac{\pi}{6} = \sqrt{3}, \quad y = 2 \sin \frac{\pi}{6} = 1$$

so

$$A = (\sqrt{3}, 1)$$

The line  $\theta = \frac{\pi}{2}$  is the  $y$ -axis, so the line  $AN$ , perpendicular to it, is horizontal. Since it passes through  $A$ , its equation is

$$y = 1$$

Now consider the shaded region  $R$ . For angles

$$\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$$

the outer boundary is the curve

$$r = 1 + 2 \sin \theta$$

and the inner boundary is the line  $y = 1$ .

Using  $y = r \sin \theta$ , the line  $y = 1$  becomes

$$r \sin \theta = 1$$

so

$$r = \frac{1}{\sin \theta} = \operatorname{cosec} \theta$$

Using the polar area formula,

$$\text{Area}(R) = \frac{1}{2} \int_{\pi/6}^{\pi/2} \left[ (1 + 2 \sin \theta)^2 - \operatorname{cosec}^2 \theta \right] d\theta$$

Expand the bracket:

$$\text{Area}(R) = \frac{1}{2} \int_{\pi/6}^{\pi/2} (1 + 4 \sin \theta + 4 \sin^2 \theta - \operatorname{cosec}^2 \theta) d\theta$$

Now integrate term by term. Using

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

we get

$$\begin{aligned} \int (1 + 4 \sin \theta + 4 \sin^2 \theta - \operatorname{cosec}^2 \theta) d\theta &= \int (1 + 4 \sin \theta + 2 - 2 \cos 2\theta - \operatorname{cosec}^2 \theta) d\theta \\ &= 3\theta - 4 \cos \theta - \sin 2\theta + \cot \theta \end{aligned}$$

So

$$\text{Area}(R) = \frac{1}{2} [3\theta - 4 \cos \theta - \sin 2\theta + \cot \theta]_{\pi/6}^{\pi/2}$$

Now substitute the limits.

At  $\theta = \frac{\pi}{2}$ :

$$3 \left( \frac{\pi}{2} \right) - 4 \cos \frac{\pi}{2} - \sin \pi + \cot \frac{\pi}{2} = \frac{3\pi}{2}$$

At  $\theta = \frac{\pi}{6}$ :

$$\begin{aligned} 3 \left( \frac{\pi}{6} \right) - 4 \cos \frac{\pi}{6} - \sin \frac{\pi}{3} + \cot \frac{\pi}{6} &= \frac{\pi}{2} - 4 \left( \frac{\sqrt{3}}{2} \right) - \frac{\sqrt{3}}{2} + \sqrt{3} \\ &= \frac{\pi}{2} - 2\sqrt{3} - \frac{\sqrt{3}}{2} + \sqrt{3} \\ &= \frac{\pi}{2} - \frac{3\sqrt{3}}{2} \end{aligned}$$

Therefore

$$\begin{aligned}\text{Area}(R) &= \frac{1}{2} \left( \frac{3\pi}{2} - \left( \frac{\pi}{2} - \frac{3\sqrt{3}}{2} \right) \right) \\ &= \frac{1}{2} \left( \pi + \frac{3\sqrt{3}}{2} \right) \\ &= \frac{\pi}{2} + \frac{3\sqrt{3}}{4}\end{aligned}$$

4. The curve  $C$  has polar equation

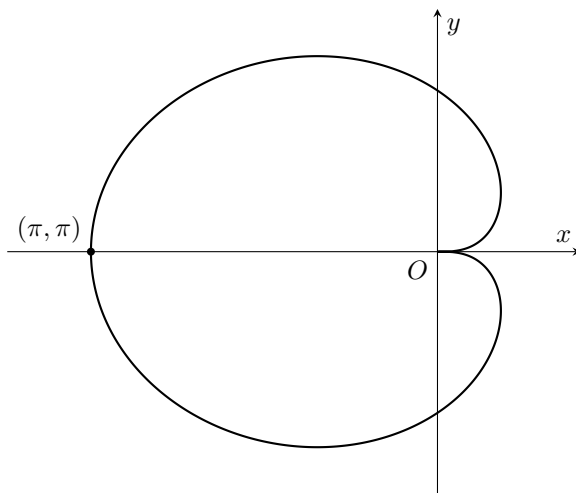
$$r^2 = \frac{\pi^2 - (\theta - \pi)^2}{1 + (\theta - \pi)^2} \quad 0 \leq \theta \leq 2\pi$$

(a) Sketch  $C$  and state the polar coordinates of the point on  $C$  furthest from the pole. [4]

(b) Using the substitution  $u = \theta - \pi$ , or otherwise, find the exact area enclosed by  $C$ . [6]

### Solution

(a) The sketch should look like:



When  $\theta = 0$  or  $\theta = 2\pi$ ,  $r^2 = \frac{\pi^2 - \pi^2}{1 + \pi^2} = 0$ , so  $C$  starts and ends at the pole. Since  $r$  depends only on  $(\theta - \pi)^2$ ,  $r(2\pi - \theta) = r(\theta)$ , so  $C$  is symmetric about the initial line.

For the furthest point, let

$$w = (\theta - \pi)^2$$

Then

$$r^2 = \frac{\pi^2 - w}{1 + w}, \quad 0 \leq w \leq \pi^2$$

Differentiate with respect to  $w$ :

$$\frac{d(r^2)}{dw} = \frac{-(1+w) - (\pi^2 - w)}{(1+w)^2} = -\frac{\pi^2 + 1}{(1+w)^2} < 0$$

So  $r^2$  decreases as  $w$  increases, and its maximum occurs when  $w = 0$ .

Hence

$$(\theta - \pi)^2 = 0 \implies \theta = \pi$$

and then

$$r^2 = \frac{\pi^2}{1} = \pi^2$$

Since we take  $r > 0$ , this gives

$$r = \pi$$

So the point on  $C$  furthest from the pole is

$$(\pi, \pi)$$

(b) The area enclosed by a polar curve is

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 \, d\theta$$

So here

$$A = \frac{1}{2} \int_0^{2\pi} \frac{\pi^2 - (\theta - \pi)^2}{1 + (\theta - \pi)^2} \, d\theta$$

Use the substitution

$$u = \theta - \pi$$

Then  $du = d\theta$ , and the limits change as follows:

$$\theta = 0 \implies u = -\pi, \quad \theta = 2\pi \implies u = \pi$$

Hence

$$A = \frac{1}{2} \int_{-\pi}^{\pi} \frac{\pi^2 - u^2}{1 + u^2} \, du$$

Now simplify the integrand:

$$\begin{aligned} \frac{\pi^2 - u^2}{1 + u^2} &= \frac{-(1 + u^2) + (\pi^2 + 1)}{1 + u^2} \\ &= -1 + \frac{\pi^2 + 1}{1 + u^2} \end{aligned}$$

Therefore

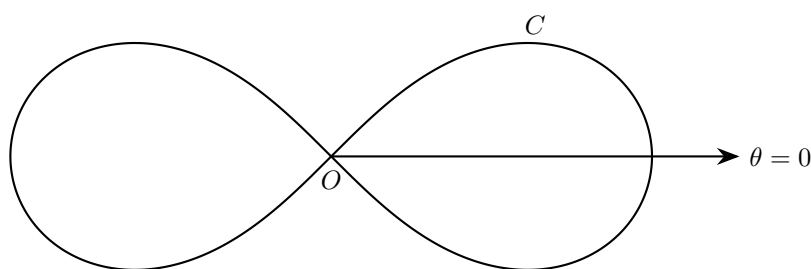
$$\begin{aligned} A &= \frac{1}{2} \int_{-\pi}^{\pi} \left( -1 + \frac{\pi^2 + 1}{1 + u^2} \right) \, du \\ &= \frac{1}{2} \left[ -u + (\pi^2 + 1) \arctan u \right]_{-\pi}^{\pi} \end{aligned}$$

using

$$\int \frac{1}{1 + u^2} \, du = \arctan u + c$$

Now substitute the limits:

$$\begin{aligned} A &= \frac{1}{2} \left[ (-\pi + (\pi^2 + 1) \arctan \pi) - (\pi - (\pi^2 + 1) \arctan \pi) \right] \\ &= \frac{1}{2} \left[ -2\pi + 2(\pi^2 + 1) \arctan \pi \right] \\ &= (\pi^2 + 1) \arctan(\pi) - \pi \end{aligned}$$



5. The diagram above shows the curve  $C$  with polar equation  $r^2 = 8 \cos 2\theta$ .

(a) Show that a cartesian equation of  $C$  is

$$(x^2 + y^2)^2 = 8(x^2 - y^2) \quad [4]$$

(b) Show that the coordinate axes are lines of symmetry of  $C$ . [2]

(c) Find the exact area enclosed by each loop of  $C$ . [5]

### Solution

(a) Using

$$x = r \cos \theta, \quad y = r \sin \theta$$

we have

$$r^2 = x^2 + y^2$$

Also,

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

and since

$$\cos \theta = \frac{x}{r}, \quad \sin \theta = \frac{y}{r}$$

it follows that

$$\begin{aligned} \cos 2\theta &= \left(\frac{x}{r}\right)^2 - \left(\frac{y}{r}\right)^2 \\ &= \frac{x^2 - y^2}{r^2} \end{aligned}$$

Now substitute into the polar equation  $r^2 = 8 \cos 2\theta$ :

$$r^2 = 8 \cdot \frac{x^2 - y^2}{r^2}$$

and since  $r^2 = x^2 + y^2$ ,

$$x^2 + y^2 = \frac{8(x^2 - y^2)}{x^2 + y^2}$$

Multiplying both sides by  $x^2 + y^2$  gives

$$(x^2 + y^2)^2 = 8(x^2 - y^2)$$

(b) To test symmetry in the  $x$ -axis, replace  $y$  by  $-y$ :

$$\begin{aligned} (x^2 + (-y)^2)^2 &= 8(x^2 - (-y)^2) \\ (x^2 + y^2)^2 &= 8(x^2 - y^2) \end{aligned}$$

which is unchanged, so the curve is symmetric about the  $x$ -axis.

To test symmetry in the  $y$ -axis, replace  $x$  by  $-x$ :

$$\begin{aligned}((-x)^2 + y^2)^2 &= 8((-x)^2 - y^2) \\ (x^2 + y^2)^2 &= 8(x^2 - y^2)\end{aligned}$$

which is also unchanged, so the curve is symmetric about the  $y$ -axis.

Hence the coordinate axes are lines of symmetry of  $C$ .

(c) For a polar curve, the area enclosed is

$$A = \frac{1}{2} \int r^2 \, d\theta$$

Here

$$r^2 = 8 \cos 2\theta$$

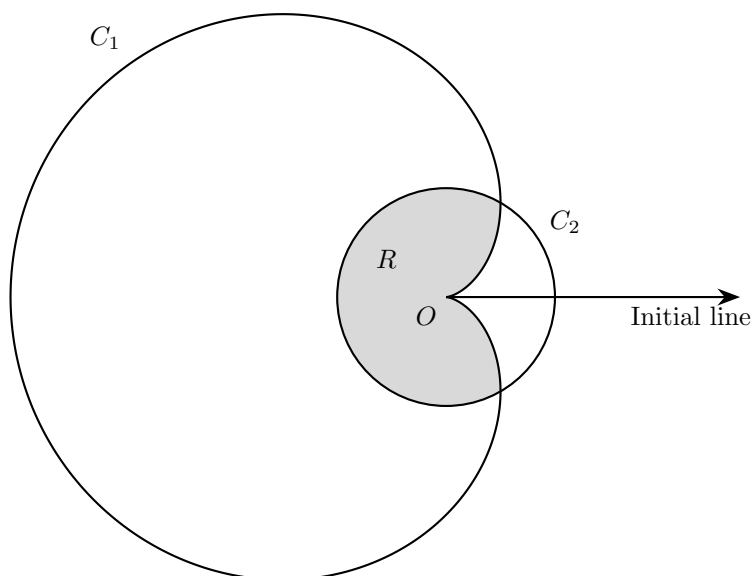
One loop is traced when  $r^2 \geq 0$  and  $\theta$  runs from  $-\frac{\pi}{4}$  to  $\frac{\pi}{4}$ , since at the endpoints

$$\cos 2\theta = 0$$

so the curve passes through the pole.

Therefore the area of one loop is

$$\begin{aligned}A &= \frac{1}{2} \int_{-\pi/4}^{\pi/4} 8 \cos 2\theta \, d\theta \\ &= 4 \int_{-\pi/4}^{\pi/4} \cos 2\theta \, d\theta \\ &= 4 \left[ \frac{1}{2} \sin 2\theta \right]_{-\pi/4}^{\pi/4} \\ &= 2 [\sin 2\theta]_{-\pi/4}^{\pi/4} \\ &= 2 \left( \sin \frac{\pi}{2} - \sin \left( -\frac{\pi}{2} \right) \right) \\ &= 2(1 - (-1)) \\ &= 4\end{aligned}$$



6. The diagram above shows the curve  $C_1$  with polar equation  $r = 2a(1 - \cos \theta)$ ,  $0 \leq \theta \leq 2\pi$ , and the circle  $C_2$  with polar equation  $r = a$ ,  $0 \leq \theta \leq 2\pi$ , where  $a$  is a positive constant.

(a) Find, in terms of  $a$ , the polar coordinates of the points where  $C_1$  meets  $C_2$ .

[3]

The common region enclosed by  $C_1$  and  $C_2$  is shaded in the diagram above.

(b) Find the area of the shaded region  $R$ , giving your answer in the form

$$\frac{1}{12}a^2 (p\pi + q\sqrt{3})$$

where  $p$  and  $q$  are integers to be found.

[7]

### Solution

(a) At the points of intersection, the two values of  $r$  are equal, so

$$2a(1 - \cos \theta) = a$$

Since  $a > 0$ , divide through by  $a$ :

$$2(1 - \cos \theta) = 1$$

$$2 - 2\cos \theta = 1$$

$$\cos \theta = \frac{1}{2}$$

For  $0 \leq \theta \leq 2\pi$ , this gives

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

Substituting into either curve gives

$$r = a$$

So the points where  $C_1$  meets  $C_2$  are

$$(a, \frac{\pi}{3}) \quad \text{and} \quad (a, \frac{5\pi}{3})$$

(b) For the common region, the boundary is given by the smaller of the two radii.

Compare the curves:

$$2a(1 - \cos \theta) \leq a$$

$$2(1 - \cos \theta) \leq 1$$

$$\cos \theta \geq \frac{1}{2}$$

So:

$$0 \leq \theta \leq \frac{\pi}{3} \quad \text{and} \quad \frac{5\pi}{3} \leq \theta \leq 2\pi$$

give the cardioid  $r = 2a(1 - \cos \theta)$  as the inner curve, and

$$\frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$$

gives the circle  $r = a$  as the inner curve.

Using the polar area formula

$$A = \frac{1}{2} \int r^2 d\theta$$

the shaded area is

$$A = \frac{1}{2} \left[ \int_0^{\pi/3} (2a(1 - \cos \theta))^2 d\theta + \int_{\pi/3}^{5\pi/3} a^2 d\theta + \int_{5\pi/3}^{2\pi} (2a(1 - \cos \theta))^2 d\theta \right]$$

By symmetry, the first and third integrals are equal, so

$$\begin{aligned} A &= \int_0^{\pi/3} (2a(1 - \cos \theta))^2 d\theta + \frac{1}{2} \int_{\pi/3}^{5\pi/3} a^2 d\theta \\ &= 4a^2 \int_0^{\pi/3} (1 - 2\cos \theta + \cos^2 \theta) d\theta + \frac{1}{2} a^2 \left( \frac{5\pi}{3} - \frac{\pi}{3} \right) \end{aligned}$$

Now use

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

So

$$\begin{aligned} A &= 4a^2 \int_0^{\pi/3} \left( 1 - 2\cos \theta + \frac{1 + \cos 2\theta}{2} \right) d\theta + \frac{1}{2} a^2 \cdot \frac{4\pi}{3} \\ &= 4a^2 \int_0^{\pi/3} \left( \frac{3}{2} - 2\cos \theta + \frac{1}{2} \cos 2\theta \right) d\theta + \frac{2\pi a^2}{3} \end{aligned}$$

Integrating,

$$\begin{aligned} A &= 4a^2 \left[ \frac{3}{2} \theta - 2 \sin \theta + \frac{1}{4} \sin 2\theta \right]_0^{\pi/3} + \frac{2\pi a^2}{3} \\ &= 4a^2 \left( \frac{3}{2} \cdot \frac{\pi}{3} - 2 \sin \frac{\pi}{3} + \frac{1}{4} \sin \frac{2\pi}{3} \right) + \frac{2\pi a^2}{3} \\ &= 4a^2 \left( \frac{\pi}{2} - 2 \cdot \frac{\sqrt{3}}{2} + \frac{1}{4} \cdot \frac{\sqrt{3}}{2} \right) + \frac{2\pi a^2}{3} \\ &= 4a^2 \left( \frac{\pi}{2} - \sqrt{3} + \frac{\sqrt{3}}{8} \right) + \frac{2\pi a^2}{3} \\ &= 4a^2 \left( \frac{\pi}{2} - \frac{7\sqrt{3}}{8} \right) + \frac{2\pi a^2}{3} \\ &= 2\pi a^2 - \frac{7\sqrt{3}}{2} a^2 + \frac{2\pi a^2}{3} \\ &= \frac{8\pi a^2}{3} - \frac{7\sqrt{3} a^2}{2} \end{aligned}$$

Putting this over a denominator of 12,

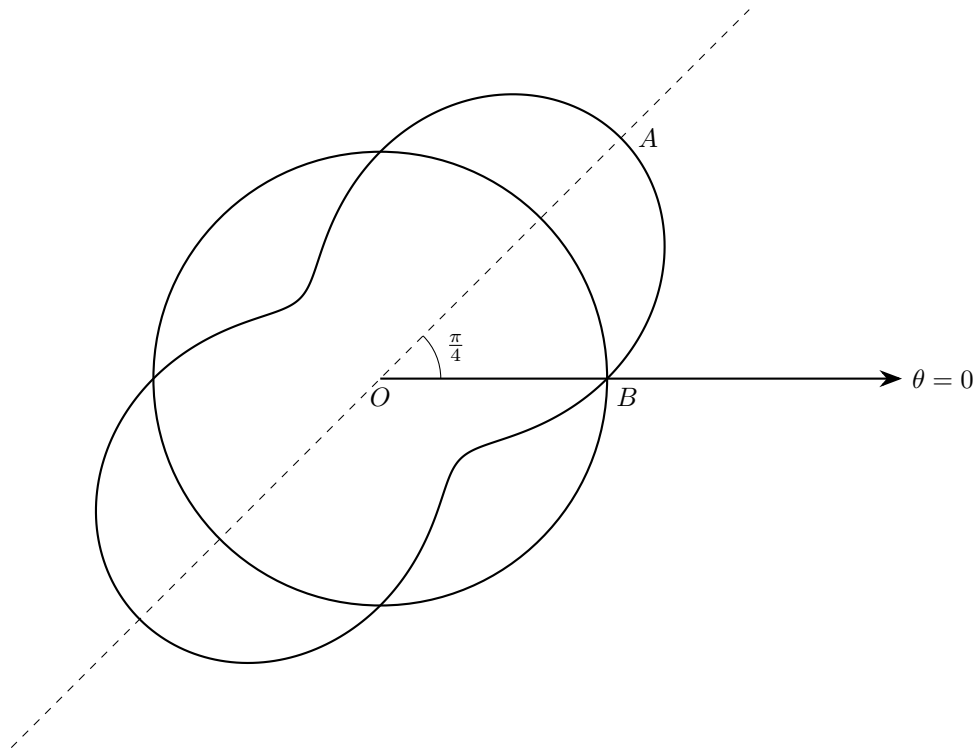
$$A = \frac{1}{12}a^2(32\pi - 42\sqrt{3})$$

Hence the shaded area is

$$\frac{1}{12}a^2(32\pi - 42\sqrt{3})$$

so

$$p = 32, \quad q = -42$$



7. The diagram above shows the curve  $C$  with polar equation  $r = a(2 + \sin 2\theta)$  and the circle  $r = 2a$  for  $-\pi \leq \theta \leq \pi$ , where  $a$  is a positive constant.

- (a) Write down the polar coordinates of the points  $A$  and  $B$ . [2]
- (b) Explain why  $C$  is symmetrical about the line  $\theta = \frac{\pi}{4}$ . [2]
- (c) Find, in terms of  $a$ , the exact total area of the regions that lie inside  $C$  and outside the circle. [5]

### Solution

- (a) For point  $A$ , the diagram shows  $\theta = \frac{\pi}{4}$ . Then

$$r = a \left( 2 + \sin \frac{\pi}{2} \right) = a(2 + 1) = 3a$$

so

$$A = \left( 3a, \frac{\pi}{4} \right)$$

For point  $B$ ,  $\theta = 0$ , so

$$r = a(2 + \sin 0) = 2a$$

hence

$$B = (2a, 0)$$

- (b) A curve is symmetrical about the line  $\theta = \frac{\pi}{4}$  if replacing  $\theta$  by

$$\frac{\pi}{2} - \theta$$

leaves the equation unchanged, since reflection in the line  $\theta = \frac{\pi}{4}$  sends  $\theta$  to  $\frac{\pi}{2} - \theta$ .

Here,

$$r = a \left( 2 + \sin 2 \left( \frac{\pi}{2} - \theta \right) \right) = a(2 + \sin(\pi - 2\theta))$$

and since  $\sin(\pi - 2\theta) = \sin 2\theta$ ,

$$r = a(2 + \sin 2\theta)$$

So the equation is unchanged, and therefore  $C$  is symmetrical about the line  $\theta = \frac{\pi}{4}$ .

(c) The region is inside  $C$  and outside the circle  $r = 2a$ , so we need

$$a(2 + \sin 2\theta) \geq 2a$$

Since  $a > 0$ , this gives

$$2 + \sin 2\theta \geq 2$$

$$\sin 2\theta \geq 0$$

On  $-\pi \leq \theta \leq \pi$ , this happens on

$$-\pi \leq \theta \leq -\frac{\pi}{2} \quad \text{and} \quad 0 \leq \theta \leq \frac{\pi}{2}$$

so there are two equal regions.

Find the area of one region, for  $0 \leq \theta \leq \frac{\pi}{2}$ .

Using the polar area formula,

$$\text{Area} = \frac{1}{2} \int r^2 d\theta$$

the required area for one region is

$$A_1 = \frac{1}{2} \int_0^{\pi/2} (a^2(2 + \sin 2\theta)^2 - (2a)^2) d\theta$$

$$A_1 = \frac{a^2}{2} \int_0^{\pi/2} ((2 + \sin 2\theta)^2 - 4) d\theta$$

Expand the bracket:

$$(2 + \sin 2\theta)^2 - 4 = 4 + 4 \sin 2\theta + \sin^2 2\theta - 4 = 4 \sin 2\theta + \sin^2 2\theta$$

So

$$A_1 = \frac{a^2}{2} \int_0^{\pi/2} (4 \sin 2\theta + \sin^2 2\theta) d\theta$$

Now evaluate each part:

$$\int_0^{\pi/2} 4 \sin 2\theta d\theta = [-2 \cos 2\theta]_0^{\pi/2} = 2 - (-2) = 4$$

Also,

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

so

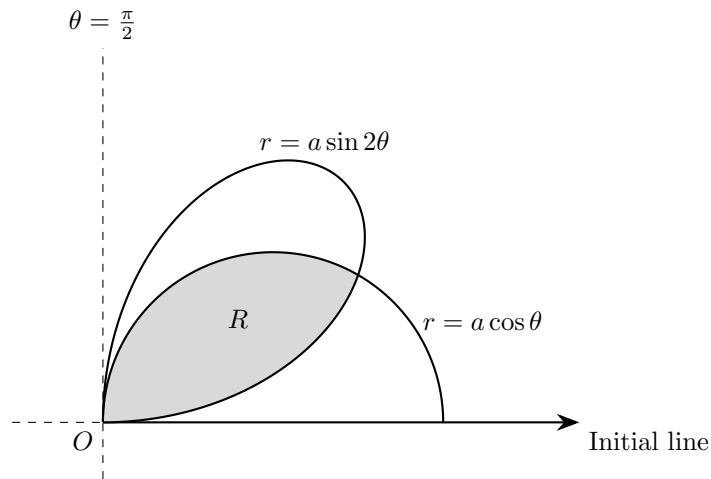
$$\int_0^{\pi/2} \sin^2 2\theta d\theta = \int_0^{\pi/2} \frac{1 - \cos 4\theta}{2} d\theta = \left[ \frac{\theta}{2} - \frac{\sin 4\theta}{8} \right]_0^{\pi/2} = \frac{\pi}{4}$$

Therefore

$$A_1 = \frac{a^2}{2} \left( 4 + \frac{\pi}{4} \right) = a^2 \left( 2 + \frac{\pi}{8} \right)$$

There are two such regions, so the total area is

$$2A_1 = 2a^2 \left( 2 + \frac{\pi}{8} \right) = a^2 \left( 4 + \frac{\pi}{4} \right) = \frac{a^2(16 + \pi)}{4}$$



8. In the first quadrant, the curve  $r = a \sin 2\theta$  intersects the circle  $r = a \cos \theta$  at the pole and at one other point.

The region  $R$  shown consists of the points that lie inside both curves.

Find the exact area of  $R$  in terms of  $a$ .

[8]

### Solution

For the points of intersection, set the two expressions for  $r$  equal:

$$a \sin 2\theta = a \cos \theta$$

Since  $a \neq 0$ ,

$$\sin 2\theta = \cos \theta$$

Using  $\sin 2\theta = 2 \sin \theta \cos \theta$ ,

$$2 \sin \theta \cos \theta = \cos \theta$$

$$\cos \theta (2 \sin \theta - 1) = 0$$

In the first quadrant,  $0 \leq \theta \leq \frac{\pi}{2}$ , so

$$\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

or

$$2 \sin \theta - 1 = 0 \Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

At  $\theta = \frac{\pi}{2}$ ,

$$r = a \cos \frac{\pi}{2} = 0$$

so this is the pole.

At the other intersection,

$$r = a \cos \frac{\pi}{6} = \frac{a\sqrt{3}}{2}$$

So the non-pole intersection is at

$$(r, \theta) = \left( \frac{a\sqrt{3}}{2}, \frac{\pi}{6} \right)$$

Now find which curve forms the boundary of the region inside both curves.

For  $0 \leq \theta \leq \frac{\pi}{6}$ ,

$$a \sin 2\theta \leq a \cos \theta$$

so the smaller radius is  $r = a \sin 2\theta$ .

For  $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$ ,

$$a \cos \theta \leq a \sin 2\theta$$

so the smaller radius is  $r = a \cos \theta$ .

Hence, using the polar area formula

$$\text{Area} = \frac{1}{2} \int r^2 d\theta$$

the required area is

$$\text{Area}(R) = \frac{1}{2} a^2 \left( \int_0^{\pi/6} \sin^2 2\theta d\theta + \int_{\pi/6}^{\pi/2} \cos^2 \theta d\theta \right)$$

Now evaluate the integrals.

First, use

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

Then

$$\begin{aligned} \int_0^{\pi/6} \sin^2 2\theta d\theta &= \int_0^{\pi/6} \frac{1 - \cos 4\theta}{2} d\theta \\ &= \left[ \frac{\theta}{2} - \frac{\sin 4\theta}{8} \right]_0^{\pi/6} \\ &= \frac{\pi}{12} - \frac{\sin(2\pi/3)}{8} \\ &= \frac{\pi}{12} - \frac{\sqrt{3}}{16} \end{aligned}$$

Next, use

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

Then

$$\begin{aligned} \int_{\pi/6}^{\pi/2} \cos^2 \theta d\theta &= \int_{\pi/6}^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta \\ &= \left[ \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_{\pi/6}^{\pi/2} \\ &= \left( \frac{\pi}{4} + 0 \right) - \left( \frac{\pi}{12} + \frac{\sin(\pi/3)}{4} \right) \\ &= \frac{\pi}{6} - \frac{\sqrt{3}}{8} \end{aligned}$$

Substitute back:

$$\begin{aligned} \text{Area}(R) &= \frac{1}{2} a^2 \left( \frac{\pi}{12} - \frac{\sqrt{3}}{16} + \frac{\pi}{6} - \frac{\sqrt{3}}{8} \right) \\ &= \frac{1}{2} a^2 \left( \frac{\pi}{4} - \frac{3\sqrt{3}}{16} \right) \\ &= \frac{a^2(4\pi - 3\sqrt{3})}{32} \end{aligned}$$

9. (a) On the same diagram, sketch the curve  $C_1$  with polar equation

$$r = 2(1 + \cos \theta), \quad 0 \leq \theta \leq 2\pi$$

and the curve  $C_2$  with polar equation  $\theta = \frac{2\pi}{3}$ .

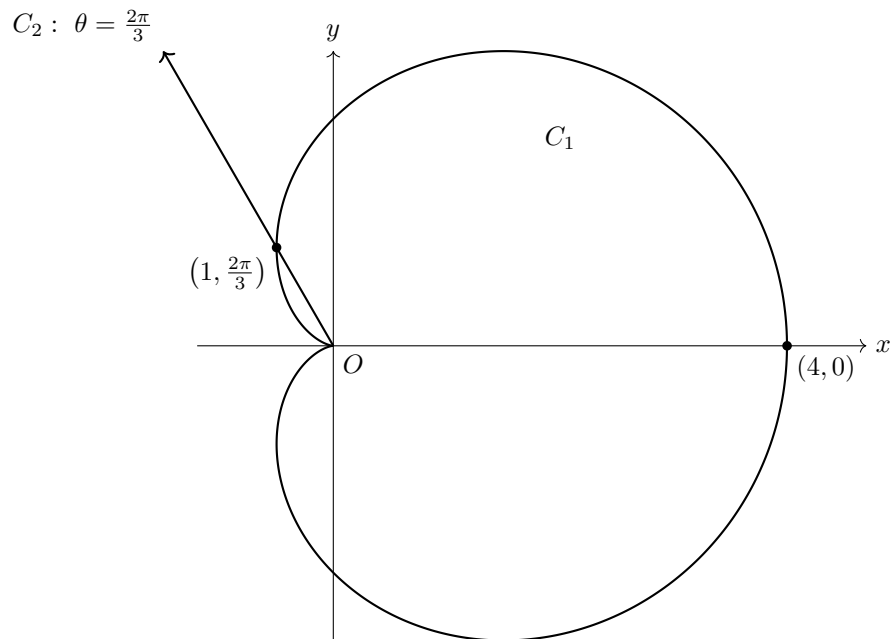
[3]

- (b) Find the area of the smaller region bounded by  $C_1$  and  $C_2$ .

[6]

### Solution

- (a) The sketch should look like:



$C_1$  is a cardioid, symmetric about the initial line, with  $r = 4$  at  $\theta = 0$  and a cusp at the pole at  $\theta = \pi$ .  $C_2$  is the half-line from the pole at angle  $\frac{2\pi}{3}$  to the initial line. Substituting  $\theta = \frac{2\pi}{3}$  into  $C_1$ :

$$r = 2 \left( 1 + \cos \frac{2\pi}{3} \right) = 2 \left( 1 - \frac{1}{2} \right) = 1$$

So the curves meet at the pole and at  $(1, \frac{2\pi}{3})$ .

- (b) The smaller region is the part inside the cardioid between

$$\theta = \frac{2\pi}{3} \quad \text{and} \quad \theta = \pi$$

Using the polar area formula

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

we get

$$A = \frac{1}{2} \int_{2\pi/3}^{\pi} [2(1 + \cos \theta)]^2 d\theta$$

Now simplify the integrand:

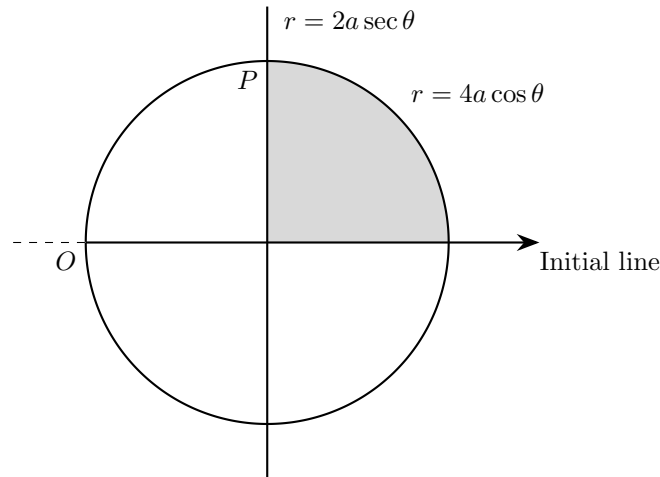
$$\begin{aligned} A &= 2 \int_{2\pi/3}^{\pi} (1 + 2\cos \theta + \cos^2 \theta) d\theta \\ &= 2 \int_{2\pi/3}^{\pi} \left( 1 + 2\cos \theta + \frac{1 + \cos 2\theta}{2} \right) d\theta \\ &= \int_{2\pi/3}^{\pi} (3 + 4\cos \theta + \cos 2\theta) d\theta \end{aligned}$$

Integrating,

$$\begin{aligned} A &= \left[ 3\theta + 4 \sin \theta + \frac{1}{2} \sin 2\theta \right]_{2\pi/3}^{\pi} \\ &= \left( 3\pi + 4 \sin \pi + \frac{1}{2} \sin 2\pi \right) - \left( 3 \cdot \frac{2\pi}{3} + 4 \sin \frac{2\pi}{3} + \frac{1}{2} \sin \frac{4\pi}{3} \right) \\ &= 3\pi - \left( 2\pi + 4 \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \left( -\frac{\sqrt{3}}{2} \right) \right) \\ &= \pi - 2\sqrt{3} + \frac{\sqrt{3}}{4} \\ &= \pi - \frac{7\sqrt{3}}{4} \end{aligned}$$

So the area of the smaller region is

$$\pi - \frac{7\sqrt{3}}{4} \approx 0.111$$



10. The diagram shows a sketch of the curves with polar equations

$$r = 4a \cos \theta \text{ and } r = 2a \sec \theta, \quad a > 0$$

- (a) Find the polar coordinates of the point of intersection  $P$  of the two curves in the first quadrant. [4]
- (b) Find the area, shaded in the figure, bounded by the two curves and by the initial line  $\theta = 0$ , giving your answer in terms of  $a$  and  $\pi$ . [6]

### Solution

- (a) At a point of intersection, the two expressions for  $r$  are equal:

$$4a \cos \theta = 2a \sec \theta$$

Since  $a > 0$ , divide by  $2a$ :

$$2 \cos \theta = \sec \theta = \frac{1}{\cos \theta}$$

So

$$2 \cos^2 \theta = 1$$

$$\cos^2 \theta = \frac{1}{2}$$

In the first quadrant,  $\cos \theta > 0$ , so

$$\cos \theta = \frac{1}{\sqrt{2}}$$

hence

$$\theta = \frac{\pi}{4}$$

Now find  $r$ :

$$r = 4a \cos \frac{\pi}{4} = 4a \cdot \frac{1}{\sqrt{2}} = 2\sqrt{2}a$$

So the polar coordinates of  $P$  are

$$\left(2\sqrt{2}a, \frac{\pi}{4}\right)$$

- (b) The required shaded area is between the curves from  $\theta = 0$  to the intersection at  $\theta = \frac{\pi}{4}$ .  
For  $0 \leq \theta \leq \frac{\pi}{4}$ , the outer curve is

$$r = 4a \cos \theta$$

and the inner curve is

$$r = 2a \sec \theta$$

So, using

$$\text{Area} = \frac{1}{2} \int_{\alpha}^{\beta} (r_{\text{outer}}^2 - r_{\text{inner}}^2) \, d\theta$$

we get

$$A = \frac{1}{2} \int_0^{\pi/4} ((4a \cos \theta)^2 - (2a \sec \theta)^2) \, d\theta$$

Simplify:

$$A = \frac{1}{2} \int_0^{\pi/4} (16a^2 \cos^2 \theta - 4a^2 \sec^2 \theta) \, d\theta$$

$$A = 8a^2 \int_0^{\pi/4} \cos^2 \theta \, d\theta - 2a^2 \int_0^{\pi/4} \sec^2 \theta \, d\theta$$

Use

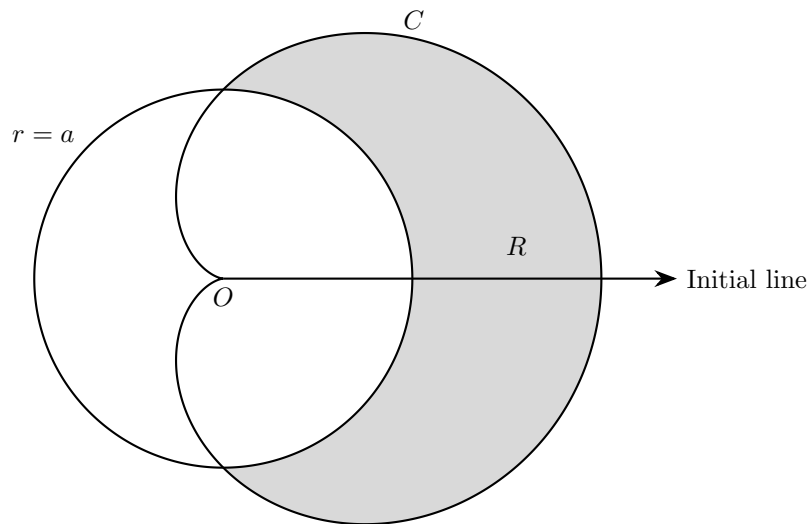
$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

so

$$\int \cos^2 \theta \, d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4}$$

Therefore

$$\begin{aligned} A &= 8a^2 \left[ \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{\pi/4} - 2a^2 [\tan \theta]_0^{\pi/4} \\ &= 8a^2 \left( \frac{\pi}{8} + \frac{1}{4} \right) - 2a^2(1) \\ &= \pi a^2 + 2a^2 - 2a^2 \\ &= \pi a^2 \end{aligned}$$



11. The figure above shows a sketch of the curve  $C$  with polar equation

$$r = a(1 + \cos \theta)$$

and the circle

$$r = a$$

where  $a > 0$  and  $-\pi \leq \theta \leq \pi$ .

The shaded region inside  $C$  and outside the circle has area

$$18 + \frac{9\pi}{4}$$

Find the value of  $a$ .

[8]

### Solution

For the shaded region, we first find where the curve and the circle meet.

The curve is

$$r = a(1 + \cos \theta)$$

and the circle is

$$r = a$$

At points of intersection,

$$a(1 + \cos \theta) = a$$

Since  $a > 0$ , divide by  $a$ :

$$1 + \cos \theta = 1$$

so

$$\cos \theta = 0$$

Hence

$$\theta = \pm \frac{\pi}{2}$$

So the shaded region is between  $\theta = -\frac{\pi}{2}$  and  $\theta = \frac{\pi}{2}$ , with outer radius

$$r = a(1 + \cos \theta)$$

and inner radius

$$r = a$$

Using the polar area formula,

$$\text{Area} = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

the shaded area is

$$\frac{1}{2} \int_{-\pi/2}^{\pi/2} ([a(1 + \cos \theta)]^2 - a^2) d\theta$$

Expand the bracket:

$$\begin{aligned} [a(1 + \cos \theta)]^2 - a^2 &= a^2(1 + 2 \cos \theta + \cos^2 \theta) - a^2 \\ &= a^2(2 \cos \theta + \cos^2 \theta) \end{aligned}$$

So

$$\text{Area} = \frac{a^2}{2} \int_{-\pi/2}^{\pi/2} (2 \cos \theta + \cos^2 \theta) d\theta$$

Now evaluate the two parts separately.

First,

$$\int_{-\pi/2}^{\pi/2} 2 \cos \theta d\theta = 2[\sin \theta]_{-\pi/2}^{\pi/2} = 2(1 - (-1)) = 4$$

For the second part, use

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

So

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta &= \int_{-\pi/2}^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta \\ &= \left[ \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_{-\pi/2}^{\pi/2} \\ &= \left( \frac{\pi}{4} + 0 \right) - \left( -\frac{\pi}{4} + 0 \right) \\ &= \frac{\pi}{2} \end{aligned}$$

Therefore

$$\begin{aligned} \text{Area} &= \frac{a^2}{2} \left( 4 + \frac{\pi}{2} \right) \\ &= 2a^2 + \frac{\pi a^2}{4} \end{aligned}$$

We are told this area is

$$18 + \frac{9\pi}{4}$$

So

$$2a^2 + \frac{\pi a^2}{4} = 18 + \frac{9\pi}{4}$$

Multiply through by 4:

$$\begin{aligned} 8a^2 + \pi a^2 &= 72 + 9\pi \\ a^2(8 + \pi) &= 9(8 + \pi) \end{aligned}$$

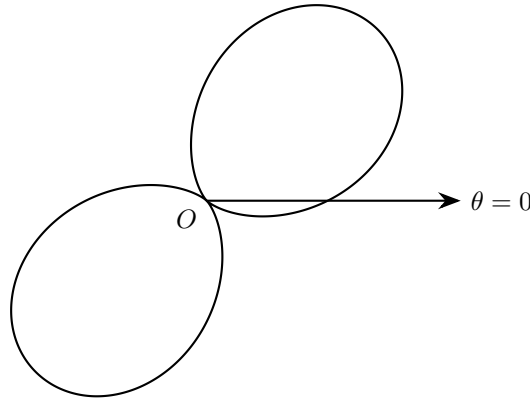
Hence

$$a^2 = 9$$

Since  $a > 0$ ,

$$a = 3$$

Therefore, the value of  $a$  is 3.



12. The diagram above shows the curve with polar equation  $r = 1 + \sin 2\theta$  for  $0 \leq \theta \leq 2\pi$ .

- (a) Find the values of  $\theta$  at the pole. [1]
- (b) Find the polar coordinates of the points on the curve where  $r$  takes its maximum value. [2]
- (c) Find the exact total area enclosed by the curve. [4]
- (d) Find a Cartesian equation for the curve. [2]

### Solution

(a) At the pole,  $r = 0$ . So

$$1 + \sin 2\theta = 0$$

$$\sin 2\theta = -1$$

For  $0 \leq \theta \leq 2\pi$ , this gives

$$2\theta = \frac{3\pi}{2}, \frac{7\pi}{2}$$

so

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

Therefore the values of  $\theta$  at the pole are  $\frac{3\pi}{4}$  and  $\frac{7\pi}{4}$ .

(b)  $r = 1 + \sin 2\theta$  is largest when  $\sin 2\theta = 1$ . Hence

$$r_{\max} = 1 + 1 = 2$$

Now  $\sin 2\theta = 1$  when

$$2\theta = \frac{\pi}{2}, \frac{5\pi}{2}$$

so

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

Therefore the points are

$$(2, \frac{\pi}{4}) \quad \text{and} \quad (2, \frac{5\pi}{4})$$

(c) The total area enclosed by a polar curve is

$$A = \frac{1}{2} \int_0^{2\pi} r^2 \, d\theta$$

So here

$$A = \frac{1}{2} \int_0^{2\pi} (1 + \sin 2\theta)^2 \, d\theta$$

Expanding,

$$A = \frac{1}{2} \int_0^{2\pi} (1 + 2 \sin 2\theta + \sin^2 2\theta) \, d\theta$$

Using

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

gives

$$\begin{aligned} A &= \frac{1}{2} \int_0^{2\pi} \left( 1 + 2 \sin 2\theta + \frac{1 - \cos 4\theta}{2} \right) \, d\theta \\ &= \frac{1}{2} \int_0^{2\pi} \left( \frac{3}{2} + 2 \sin 2\theta - \frac{1}{2} \cos 4\theta \right) \, d\theta \end{aligned}$$

Integrating term by term,

$$\begin{aligned} A &= \frac{1}{2} \left[ \frac{3}{2}\theta - \cos 2\theta - \frac{1}{8} \sin 4\theta \right]_0^{2\pi} \\ &= \frac{1}{2} (3\pi - 1 - 0 - (-1)) \\ &= \frac{3\pi}{2} \end{aligned}$$

(d) Using

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2$$

and  $\sin 2\theta = 2 \sin \theta \cos \theta$ ,

$$r = 1 + 2 \sin \theta \cos \theta$$

Now

$$\sin \theta = \frac{y}{r}, \quad \cos \theta = \frac{x}{r}$$

so

$$r = 1 + \frac{2xy}{r^2}$$

Multiplying by  $r^2$ ,

$$r^3 = r^2 + 2xy$$

Since  $r^2 = x^2 + y^2$ ,

$$r^3 = x^2 + y^2 + 2xy = (x + y)^2$$

Also  $r = \sqrt{x^2 + y^2}$ , so the cartesian equation is

$$(x^2 + y^2)^{3/2} = (x + y)^2$$